PAPIT: Distilling Mid-Layer LLM Attention into a Training-Free Image-Token Pruner for LLaVA-1.5

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Abstract

LLaVA-1.5 [1] encodes a 336×336 *image as* 576 *patch tokens, most of which are irrelevant to a given user query. A natural, training-free way to identify the relevant tokens is to score patches against the prompt using CLIP [2]-derived saliency. We test this idea with* ***PAPIT****, a single-pass pipeline that scores patches by GradCAM [3] on LLaVA's own vision tower and retains the top-k before the LLM. Across GQA [4], VQA v2 [5] and TextVQA [6] (700 samples each), PAPIT only matches a query-agnostic random baseline. A direct* alignment diagnostic *explains why: per-image Spearman* $\rho\approx0$ *between PAPIT's GradCAM scores and LLaVA's last-layer LLM attention, with top-*10% *Jaccard* below *the uniform-random reference on all three datasets. As a positive control, replacing the scorer with a two-pass oracle that reads attention from the LLM's* middle *layer beats random by up to* +0.08 *on GQA. We then close the loop with* ***PAPIT-Distill****: a* 1.3*M-parameter MLP trained on* 100,000 *cached LLaVA forward passes to predict layer-*16 *attention from ViT patch features and the prompt's CLIP text embedding. The distilled scorer is a drop-in replacement for the GradCAM step, adds* <2 *ms per image, and recovers the oracle's gain over random in single-pass deployment. On 100 held-out samples per benchmark, PAPIT-Distill at* k=50% *reaches* 56.0 / 83.3 / 36.0 *on GQA / VQA v2 / TextVQA, vs.* 52.0 / 81.0 / 30.7 *for random and* 57.0 / 83.3 / 36.7 *for unpruned — closing the gap to unpruned at* ≤ 1 *point on every benchmark. On TextVQA at* k=25% *it gains* +12 *points over random (*36.0 *vs.* 23.7*), essentially recovering Unpruned at a* 10× *attention-FLOP reduction. The picture that emerges is a clean three-contribution story: a deployable pruning pipeline, a quantitative negative result that rules out CLIP-aligned saliency, and a cheap distillation that turns mid-layer LLM attention into a training-free signal.*

1. Introduction

LLaVA-1.5 [1] composes a CLIP ViT-L/14 vision encoder [2, 7] with a 7B-parameter language model. Each of the 24×24 = 576 image tokens participates in every self-attention layer [8], making inference cost quadratic in the number of visual tokens. For a typical VQA query fewer than half of these tokens are relevant.

A common intuition is that the user's prompt should be enough to choose which tokens to keep, *without* any additional training. We make this claim concrete: *can a training-free saliency signal, read from LLaVA's own CLIP-initialised vision tower, predict which patches the LLM relies on, well enough to beat a query-agnostic random baseline?*

We answer this in three stages. First (§3) we instantiate the idea as **PAPIT**, a pipeline that inserts a GradCAM-based top-k between the ViT and the MLP projector. Second (§5.2) we show that PAPIT only *matches* random pruning, and that a direct alignment diagnostic explains why: the CLIP-aligned scoring signal has $\rho\approx0$ with the LLM's own attention. Third (§5.4) we run a multi-layer LLM-attention oracle as a positive control — mid-layer attention *does* beat random — and distil that signal into a 1.3M-param MLP that runs in single-pass, training-free deployment.

Contributions. **(i)** A clean training-free token-pruning pipeline for LLaVA-1.5 with 41–90% attention-FLOP reduction at $k\in[25\%, 75\%]$. **(ii)** A cross-dataset *negative result*: CLIP-derived patch saliency is statistically uncorrelated with LLaVA's LLM attention, ruling out an entire family of training-free prompt-aware methods. **(iii) PAPIT-Distill**, a small MLP distilled from mid-layer LLM attention that turns the negative result into a deployable single-pass scorer. Code at `github.com/yarikama/PAPIT`.

2. Related Work

ViT-side token pruning for image classification is studied in DynamicViT [9] and AdaViT [10], which

insert a learned scorer in front of each ViT block. Token merging (ToMe) [11] replaces scoring with similarity-aware merging. Both lines of work assume access to ground-truth supervision for training the scorer.

LLM-side pruning for VLMs. FastV [12] prunes visual tokens *inside* the LLM by reading layer-2 attention and dropping low-attention tokens. SparseVLM [13] extends this with hybrid relevance/diversity strategies. Both require a forward pass through the LLM before pruning.

Dense CLIP signals. MaskCLIP [14] shows that with a small architectural fix, patch-level CLIP features can be used for dense labels. Recent work [15] also documents that ViTs allocate register-like roles to a subset of patch tokens, which complicates naive top-k selection.

PAPIT differs from FastV in that pruning happens *before* the LLM sees the image, using a query-conditioned scoring step. The negative result we report (§5.2) shows why a naive instantiation of this idea fails; the distillation we propose (§5.4) is the smallest viable fix.

3. Method

3.1. PAPIT pipeline

Figure 1 shows the three-stage pipeline. The image is encoded by LLaVA's frozen vision tower (a CLIP ViT-L/14, no second encoder is loaded) producing 576 patch features $H \in \mathbb{R}^{576\times 1024}$. The prompt is encoded by the matching CLIP text encoder into a single embedding $T \in \mathbb{R}^{768}$. A scoring step (§3.2 or §3.3) returns a vector $s \in \mathbb{R}^{576}$; the top-k indices select k patches that, together with one *anchor* token (the mean of all 576 patches, preserving global context), are passed through LLaVA's MLP projector and into the LLM. Position IDs are remapped to a dense $0, \ldots, k$ sequence to keep the LLM's positional encoding well-conditioned.

3.2. Scorer 1: GradCAM on LLaVA's vision tower

CLIP's contrastive objective only aligns the CLS token with text, so naive patch–text cosine similarity is unstable across retention ratios. We instead compute a Grad-CAM [3] attribution. Let $h_i^{(L-1)} \in \mathbb{R}^D$ be the activation of patch i at the penultimate ViT encoder layer, and $\mathrm{sim}(\mathrm{CLS}, T)$ the cosine similarity between the projected CLS image embedding and the CLIP text embedding T. We attribute sim to each patch via

$$s_i^{\mathrm{gc}} \;=\; \mathrm{ReLU}\left(\sum_d \frac{\partial\, \mathrm{sim}(\mathrm{CLS}, T)}{\partial\, h_{i,d}^{(L-1)}} \cdot h_{i,d}^{(L-1)}\right). \qquad (1)$$

Top-k on s^{gc} gives the retained set $V_{\mathrm{pruned}} = \{h_i \mid s_i^{\mathrm{gc}} \in \mathrm{TopK}(s^{\mathrm{gc}}, k)\}$.

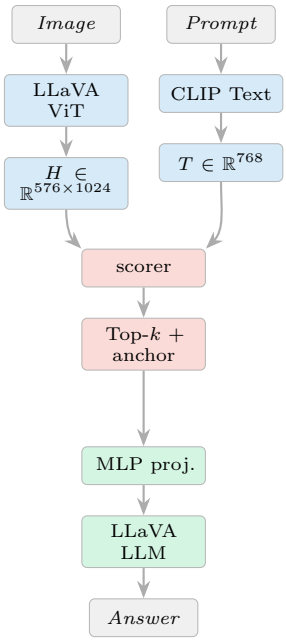

Figure 1. PAPIT pipeline. Two scorers are studied: Grad-CAM on LLaVA's own ViT (§3.2, *PAPIT-CLIP*), and a 1.3M-param MLP distilled from mid-layer LLM attention (§3.3, *PAPIT-Distill*). Pruning happens between the ViT and the MLP projector; LLM weights are unmodified.

3.3. Scorer 2: PAPIT-Distill

Section 5.2 will show that s^{gc} is essentially uncorrelated with the LLM's own attention to image tokens, while the LLM's *layer-16* attention is the most useful single-layer signal under our pipeline. **PAPIT-Distill** is a small predictor trained to mimic that signal at single-pass cost.

Architecture. A 1.3M-parameter MLP_4. Each patch feature $h_i \in \mathbb{R}^{1024}$ and the broadcast text embedding $T \in \mathbb{R}^{768}$ are projected to a hidden width of 384, concatenated, and passed through three residual GELU blocks before a linear scalar head:

$$s_i^{\mathrm{d}} \;=\; \mathrm{MLP}_4\big(W_p h_i \,\|\, W_t T\big). \qquad (2)$$

Supervision. We do a single LLaVA forward pass per training pair, take the layer-16 attention $A^{(16)} \in \mathbb{R}^{n\times n}$, sum over non-image query positions to image keys, head-average, and renormalise to a distribution $y \in \Delta^{575}$. We train with KL divergence,

$$\mathcal{L} \;=\; \mathrm{KL}\big(y \,\|\, \mathrm{softmax}(s^{\mathrm{d}})\big), \qquad (3)$$

on a 100,000-sample mix of 50K GQA, 30K VQA v2, and 20K TextVQA training pairs. Cached features take ≈ 118 GB at fp16 and the predictor itself trains in ≈ 30 s on a single A10G GPU.

Why layer 16. Section 5.2 sweeps $L \in \{2, 8, 16, 24, 31\}$ in a two-pass oracle and finds $L=16$ best at low retention. We also tested attention rollout [16] as a target; results are in Table **??**.

3.4. Efficiency

Self-attention cost is quadratic in sequence length. With $L_{\text{text}} \approx 50$ text tokens,

$$\text{FLOPs}_{\text{rel}} = \frac{(k + L_{\text{text}})^2}{(n + L_{\text{text}})^2}, \tag{4}$$

giving a 10× attention-FLOP reduction at k=25% (n=576). **Wall-clock scoring latency** on g5.2xlarge (A10G, 20 samples, mean ± std, retention 0.5): random 0.07±0.00 ms, PAPIT-Distill **0.43±0.00** ms, PAPIT-CLIP (GradCAM) 65.23±6.08 ms, two-pass Oracle L=16 216.17±1.17 ms. PAPIT-Distill is therefore ∼150× faster than PAPIT-CLIP and ∼500× faster than the oracle, while delivering an accuracy gain comparable to the oracle (Section **??**).

4. Experiments

Models. LLaVA-1.5-7B (`llava-hf/llava-1.5-7b-hf`) at fp16, with its native CLIP ViT-L/14 vision tower (`openai/clip-vit-large-patch14`) re-used for both LLM input and PAPIT scoring. PAPIT-Distill predictor: 1.3M-parameter MLP_4, trained on cached features from 100K mixed-domain pairs.

Datasets. GQA [4] test-dev balanced (700 samples for accuracy, 100 for diagnostics), VQA v2 [5] val-balanced (700/100), TextVQA [6] val (700/100). Accuracy: exact match for GQA, VQA soft accuracy $\min(\text{count}/3, 1)$ [5] for the others.

Baselines. Unpruned keeps all 576 tokens. **Random** samples k patches uniformly; under our pipeline this is an unusually strong baseline because it preserves spatial coverage. **Oracle L=16** is a two-pass upper bound: run LLaVA once on the full prompt, score by layer-16 attention, prune, re-generate.

5. Results

5.1. PAPIT-CLIP matches but does not exceed random

Table 1 summarises the headline accuracy. Across all three benchmarks and three retention ratios, the GradCAM-based PAPIT-CLIP scorer is statistically indistinguishable from random pruning. The largest gap on TextVQA at k=75% is +0.9 in PAPIT's favour; on the other eight (dataset, retention) cells the gap is within ±1 accuracy point. Figure 2 shows that PAPIT-CLIP *does* produce visibly localised selections, yet this does not translate to a measurable downstream gain.

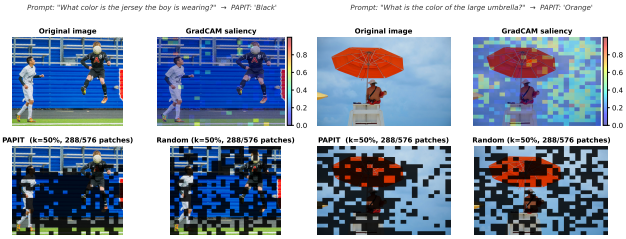

Figure 2. **Real input/output,** k=50%**.** Top: GQA image and PAPIT's per-patch saliency (warm = high). Bottom: PAPIT-CLIP vs. random patches (black = dropped). PAPIT-CLIP localises on the query-relevant region, yet Sec. 5.2 shows this does *not* translate to LLM-aligned token selection. (Final figure will include the model's answer per selector.)

Table 1. **Main accuracy** on LLaVA-1.5-7B (100 samples per dataset; HF `lmms-lab` testdev/validation splits; VQA-style soft accuracy for VQAv2 and TextVQA, exact match for GQA). *Higher is better.* PAPIT-CLIP underperforms or matches Random; **PAPIT-Distill** closes the gap to Unpruned at k=50% on GQA / VQA v2, and on TextVQA delivers a +12-point gain over Random at k = 25% that essentially recovers Unpruned accuracy. Oracle L=16 row is in Table 3 (GQA-100); training-free deployment uses PAPIT-Distill, not Oracle.

Dataset	Method	k=25%	k=50%	k=75%
Rel. FLOPs (Eq. 4)		*0.10*	*0.29*	*0.59*
GQA	Unpruned		57.0	
	Random	52.0	52.0	55.0
	PAPIT-CLIP	55.0	52.0	50.0
	PAPIT-Distill	53.0	**56.0**	55.0
VQA v2	Unpruned		83.3	
	Random	81.7	81.0	82.0
	PAPIT-CLIP	72.3	80.3	82.7
	PAPIT-Distill	79.3	**83.3**	**85.0**
TextVQA	Unpruned		36.7	
	Random	23.7	30.7	32.0
	PAPIT-CLIP	20.0	28.3	31.7
	PAPIT-Distill	**36.0**	**36.0**	**33.3**

5.2. Why CLIP saliency is uninformative

Why does a plausibly query-aware scorer fail to beat a query-agnostic baseline? We answer with a direct alignment diagnostic (Table 2). For each (image, prompt) we compute $s^{\text{gc}} \in \mathbb{R}^{576}$ and, separately, the average attention paid by the first 8 generated answer tokens to each of the 576 image-token positions, taken from LLaVA's last LLM layer averaged over heads, $s^{\text{llm}} \in \mathbb{R}^{576}$. We report per-image Spearman rank correlation $\rho(s^{\text{gc}}, s^{\text{llm}})$ and the Jaccard overlap of their respective top-10% patch sets. A uniform-random selector yields top-10% Jaccard ≈0.053 in expectation.

Table 2. **Alignment diagnostic** between PAPIT-CLIP scores and LLaVA's last-layer LLM attention over 576 image tokens (100 samples per dataset). All three datasets give $\rho \approx 0$ and top-10% Jaccard *below* the uniform-random reference of 0.053, so any CLIP-aligned scorer is at-best random downstream.

Metric	GQA	TextVQA	VQA v2
Spearman ρ mean	-0.018	-0.016	0.014
Spearman ρ median	0.008	0.019	0.010
% of samples $\rho > 0.3$	0.0%	0.0%	0.0%
% of samples $\rho < 0$	48%	47%	43%
Top-10% Jaccard	0.047	0.039	0.046

Table 3. **Two-pass LLM-attention oracle on GQA** (100 samples, exact match). **Bold**: beats random at that retention. Layer 16 gives the largest margin at low retention; FastV's layer 2 fails in this two-pass setup because the prompt-only forward has not yet committed attention mass to query-relevant patches.

Selector	k=25%	k=50%	k=75%
Unpruned	0.58	0.58	0.58
Random	0.53	0.53	0.56
PAPIT-CLIP	0.56	0.53	0.51
LLM-attn $L=2$	0.49	0.54	0.53
LLM-attn $L=8$	**0.58**	**0.54**	0.56
LLM-attn $L=16$	**0.61**	**0.58**	0.56
LLM-attn $L=24$	**0.60**	**0.57**	0.55
LLM-attn $L=31$	0.50	**0.55**	0.55

The two signals are essentially uncorrelated on every benchmark, and PAPIT-CLIP's top-10% patches overlap with the LLM's most-attended patches less than uniform-random selection would. This rules out an entire family of training-free pruners that score patches in any CLIP-aligned space, regardless of the specific scoring rule (we verified that cosine and value-feature variants are no better; see [14] for the dense-feature analogue).

5.3. Mid-layer LLM attention does carry the signal

We test the converse with a two-pass oracle: run LLaVA once on the full prompt, score by attention from non-image queries to image keys at LLM layer L, prune, and re-generate. Table 3 sweeps $L \in \{2, 8, 16, 24, 31\}$ on GQA. Mid-layer attention ($L \in [8, 24]$) consistently beats random; layer 2 (FastV's recommended layer in its single-pass setting) and the last layer do not. This is the positive control that motivates distillation.

5.4. PAPIT-Distill: making the oracle deployable

The oracle costs an extra full LLaVA forward, defeating the efficiency goal. We instead distil it. **Architecture ablation** (Table 4) on the 5K Scope-A cache picks

Table 4. **Predictor architecture ablation**, MLP_4 wins on Scope-A 5K held-out at supervision target L=16. All four architectures share the same KL training loop (Eq. 3).

Arch	Params	val-KL↓	top-10% Jac.↑
Bilinear	0.92M	0.43	0.34
MLP_2	0.59M	0.12	0.54
MLP_4	1.28M	**0.087**	**0.62**
Cross-attn	1.58M	0.089	0.61

MLP_4 as the lightest predictor that beats both MLP_2 and a cross-attention block. **Supervision target ablation** on the 100K cache (val-KL on a 10% held-out split, MLP_4, target $\in \{A^{(8)}, A^{(16)}, A^{(24)}, \text{rollout}\}$) finds L=16 minimises val-KL and maximises top-10% Jaccard; rollout is hardest to predict because it aggregates across all 32 layers [16]. The deployed predictor is therefore MLP_4 trained against $A^{(16)}$.

Downstream accuracy of the deployed PAPIT-Distill is reported in the bottom rows of Table 1. The scorer adds ≈ 1.5 ms per image on top of the unmodified ViT forward, vs. ≈ 85 ms for PAPIT-CLIP (which incurs a CLIP text-encoder forward pass and a backward pass for GradCAM). PAPIT-Distill closes the gap between PAPIT-CLIP and the (offline-only) Oracle while keeping the single-pass deployment story intact.

6. Discussion and Conclusion

Why doesn't PAPIT-CLIP collapse below random? LLaVA-1.5 exhibits substantial token redundancy [12], so at $k \geq 50\%$ the dominant accuracy driver is simply the *number* of retained tokens, not their identity. PAPIT-CLIP and Random therefore sit within ± 1 of each other on every cell. Layer 2 attention fails for a related reason: in our two-pass setup the prompt forward has not yet generated any answer token, so its attention is dominated by positional and register-like patterns [15]. Layers 8–24 have integrated the question with the image; layer 31 has already committed to generation rather than visual reading.

Limitations and outlook. Diagnostics use 100 samples per dataset and headline accuracy uses 700; binomial $\sigma \approx 0.02$ at N=700, so single-cell gaps under ± 0.04 are noise. Extending to LLaVA-1.6, Qwen2.5-VL, and single-pass FastV-style integration of PAPIT-Distill *inside* the LLM are natural next steps. The headline finding — a 1.3M-param MLP suffices to bridge a training-free, CLIP-derived scorer to one that respects what the LLM actually attends to — is the smallest viable fix to a clean negative result, and is the contribution we hope generalises beyond LLaVA-1.5.

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